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Semiconductor device and manufacturing method thereof

Abstract

A semiconductor device and its manufacturing method are provided. The semiconductor device includes a substrate, an oxygen-containing protrusive structure disposed above the substrate, a metal oxide layer, a gate dielectric layer disposed on the metal oxide layer, and a gate disposed on the gate dielectric layer. The oxygen-containing protrusive structure has a first surface, a second surface opposite to the first surface, and sidewalls connected to the first and second surfaces. The metal oxide layer includes first, second, and third portions. The first portion covers the first surface. The second portion is connected to the first portion and covers the sidewalls of the oxygen-containing protrusive structure. A resistivity of the second portion gradually decreases away from the first portion. The third portion is connected to the second portion and extends from the sidewalls of the oxygen-containing protrusive structure in a direction away from the oxygen-containing protrusive structure.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION (1) This application claims the priority benefit of U.S. provisional application Ser. No. 63/287,695, filed on Dec. 9, 2021 and Taiwan patent application serial no. 111117041, filed on May 5, 2022. The entirety of each of the above-mentioned patent applications is hereby incorporated by reference herein and made a part of this specification.

BACKGROUND
Technical Field
(1) The disclosure relates to a semiconductor device and a manufacturing method thereof; more particularly, the disclosure relates to a semiconductor device including an oxygen-containing protrusive structure and a manufacturing method thereof.
Description of Related Art
(2) Generally, a semiconductor layer of a thin film transistor (TFT) may be divided into a channel region and a doped region. If a carrier concentration of the doped region is high, and if a carrier concentration between the doped region and the channel region is suddenly dropped, a large lateral electric field may be accordingly generated near a drain of the TFT during an operation under a large current, and degradation of the semiconductor device may be induced. However, if the carrier concentration of the doped region is reduced to prevent said degradation of the semiconductor device, an operating current of the semiconductor device may be insufficient. Therefore, how to reduce the lateral electric field near the drain of the semiconductor device while maintaining sufficient operating current is an issue to be solved at present.
SUMMARY
(3) The disclosure provides a semiconductor device and a manufacturing method thereof which may reduce a lateral electric field near a drain, so as to improve reliability of the semiconductor device.
(4) In an embodiment of the disclosure, a semiconductor device that includes a substrate, an oxygen-containing protrusive structure, a metal oxide layer, a gate dielectric layer, and a first gate is provided. The oxygen-containing protrusive structure is disposed above the substrate. The oxygen-containing protrusive structure has a first surface, a second surface opposite to the first surface, and a plurality of sidewalls connected to the first surface and the second surface. The metal oxide layer includes a first portion, a second portion, and a third portion. The first portion covers the first surface of the oxygen-containing protrusive structure. The second portion is connected to the first portion and covers the sidewalls of the oxygen-containing protrusive structure. A resistivity of the second portion gradually decreases away from the first portion. The third portion is connected to the second portion and extends from the sidewalls of the oxygen-containing protrusive structure in a direction away from the oxygen-containing protrusive structure. The gate dielectric layer is disposed on the metal oxide layer. The first gate is disposed on the gate dielectric layer.
(5) In an embodiment of the disclosure, a manufacturing method of a semiconductor device includes following steps. A substrate is provided. An oxygen-containing protrusive structure is formed above the substrate, where the oxygen-containing protrusive structure has a first surface, a second surface opposite to the first surface, and a plurality of sidewalls connected to the first surface and the second surface. A metal oxide layer is formed on the oxygen-containing protrusive structure, where the metal oxide layer includes a first portion, a second portion, and a third portion. The first portion covers the first surface of the oxygen-containing protrusive structure. The second portion is connected to the first portion and covers the sidewalls of the oxygen-containing protrusive structure. The third portion is connected to the second portion and extends from the

sidewalls of the oxygen-containing protrusive structure in a direction away from the oxygen-containing protrusive structure. A gate dielectric layer is formed on the metal oxide layer. A first gate is formed on the gate dielectric layer, where the first gate is overlapped with the metal oxide layer in a normal direction of a top surface of the substrate. A doping process is performed on the metal oxide layer to reduce a resistivity of the third portion of the metal oxide layer and gradually decrease a resistivity of the second portion of the metal oxide layer away from the first portion.
(6) To make the aforementioned more comprehensible, several embodiments accompanied with drawings are described in detail as follows.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) The accompanying drawings are included to provide a further understanding of the disclosure, and are incorporated in and constitute a part of this specification. The drawings illustrate exemplary embodiments of the disclosure and, together with the description, serve to explain the principles of the disclosure.
(2) FIG. **1** is a schematic cross-sectional view of a semiconductor device according to an embodiment of the disclosure.
(3) FIG. **2**A to FIG. **2**E are schematic cross-sectional flowcharts of a manufacturing process of the semiconductor device depicted in FIG. **1**.
(4) FIG. **3** is a schematic cross-sectional view of a semiconductor device according to another embodiment of the disclosure.
(5) FIG. **4**A to FIG. **4**D are schematic cross-sectional flowcharts of a manufacturing process of the semiconductor device depicted in FIG. **3**.
(6) FIG. **5** is a schematic cross-sectional view of a semiconductor device according to still another embodiment of the disclosure.
(7) FIG. **6**A to FIG. **6**D are schematic cross-sectional flowcharts of a manufacturing process of the semiconductor device depicted in FIG. **5**.
DESCRIPTION OF THE EMBODIMENTS
(8) Reference is now made in detail to exemplary embodiments of the disclosure, and examples of the exemplary embodiments are described in the accompanying drawings. Whenever possible, the same reference numbers are used in the drawings and descriptions to indicate the same or similar parts.
(9) FIG. **1** is a schematic cross-sectional view of a semiconductor device according to an embodiment of the disclosure.
(10) With reference to FIG. **1**, a semiconductor device **1** includes a substrate **100**, an oxygen-containing protrusive structure **122**, a metal oxide layer **130**, a gate dielectric layer **140**, and a first gate **150**. In this embodiment, the semiconductor device **1** further includes a buffer layer **110**, an interlayer dielectric layer **160**, a source **172**, and a drain **174**.
(11) A material of the substrate **100** may include glass, quartz, organic polymer, or an opaque/reflective material (e.g., a conductive material, metal, wafer, ceramics, or other applicable materials), or other applicable materials. If the conductive material or the metal is used, the substrate **100** is covered by an insulation layer (not shown) to prevent short circuits. In some embodiments, the substrate **100** is a flexible substrate, and the material of the substrate **100** is, for instance, polyethylene terephthalate (PET), polyethylene naphthalate (PEN), polyester (PES), polymethylmethacrylate (PMMA), polycarbonate (PC), polyimide (PI), metal foil, or other flexible materials. The buffer layer **110** is located on the substrate **100**, and a material of the buffer layer **110** may include aluminum oxide, silicon nitride oxide (SiNO), silicon nitride, or other insulation materials, which should however not be construed as a limitation in the disclosure.

(12) The oxygen-containing protrusive structure **122** is disposed above the substrate **100** and the buffer layer **110**; that is, the buffer layer **110** is disposed between the substrate **100** and the oxygen-containing protrusive structure **122**. The oxygen-containing protrusive structure **122** has a first surface **122***a*, a second surface **122***b* opposite to the first surface **122***a*, and a plurality of sidewalls **122***c* connected to the first surface **122***a* and the second surface **122***b*. For instance, in this embodiment, the oxygen-containing protrusive structure **122** is a trapezoidal structure, the second surface **122***b* faces a surface of the buffer layer **110**, an area occupied by the first surface **122***a* is smaller than an area occupied by the second surface **122***b*, and the sidewalls **122***c* are connected to the first surface **122***a* and the second surface **122***b* to form an inclined surface. Therefore, at the sidewalls **122***c* of the oxygen-containing protrusive structure **122**, a thickness of the oxygen-containing protrusive structure **122** gradually decreases from the center to the edge. A material of the oxygen-containing protrusive structure **122** may include silicon oxide, SiNO, or other appropriate oxygen-containing insulation materials. In some embodiments, an oxygen concentration of the oxygen-containing protrusive structure **122** is higher than an oxygen concentration of the buffer layer **110**. For instance, when the material of the oxygen-containing protrusive structure **122** includes silicon oxide, and the material of the buffer layer **110** includes silicon nitride or SiNO, the oxygen concentration of the buffer layer **110** is lower than the oxygen concentration of the oxygen-containing protrusive structure **122**. Alternatively, when the materials of the buffer layer **110** and the oxygen-containing protrusive structure **122** are both SiNO, the oxygen concentration of the buffer layer **110** is lower than the oxygen concentration of the oxygen-containing protrusive structure **122**. In some embodiments, when the material of the buffer layer **110** includes silicon nitride or SiNO, the oxygen-containing protrusive structure **122** and the buffer layer **110** contain hydrogen atoms, and a hydrogen concentration of the buffer layer **110** is higher than a hydrogen concentration of the oxygen-containing protrusive structure **122**.
(13) The metal oxide layer **130** is located on the oxygen-containing protrusive structure **122** and the buffer layer **110**. For instance, the metal oxide layer **130** includes a first portion **132**, a second portion **134**, and a third portion **136**. The first portion **132** covers the first surface **122***a* of the oxygen-containing protrusive structure **122**. The second portion **134** is connected to the first portion **132** and covers the sidewalls **122***c* of the oxygen-containing protrusive structure **122**. The third portion **136** is connected to the second portion **134** and extends from the sidewalls **122***c* of the oxygen-containing protrusive structure **122** in a direction away from the oxygen-containing protrusive structure **122**. In some embodiments, a material of the metal oxide layer **130** includes quaternary metal compounds, such as indium gallium zinc oxide (IGZO), indium tin zinc oxide (ITZO), aluminum zinc tin oxide (AZTO), indium tungsten zinc oxide (IWZO), and so forth, or includes oxides composed of any three of the following ternary metals: gallium (Ga), zinc (Zn), indium (In), tin (Sn), aluminum (Al), and tungsten (W).
(14) The gate dielectric layer **140** is disposed on the metal oxide layer **130** and the buffer layer **110**, and the first gate **150** is disposed on the gate dielectric layer **140**. The first gate **150** is overlapped with the first portion **132** in a normal direction ND of a top surface of the substrate **100** but is not overlapped with the second portion **134** nor the third portion **136**. In other words, the first portion **132** of the metal oxide layer **130** may constitute a channel region ch, and the second portion **134** and the third portion **136** may constitute a doped region dp. In some embodiments, the second portion **134** may constitute a lightly doped region ldp, and the third portion **136** may constitute a heavily doped region hdp.
(15) The oxygen-containing protrusive structure **122** may provide an oxygen element to the metal oxide layer **130** in the manufacturing process. The larger the thickness of the oxygen-containing protrusive structure **122** is, the more oxygen the oxygen-containing protrusive structure may provide to the metal oxide layer **130**; the smaller the thickness of the oxygen-containing protrusive structure **122** is, the less oxygen the oxygen-containing protrusive structure may provide. The higher the oxygen concentration of the metal oxide layer **130**, the higher the resistivity of the metal

oxide layer **130**. On the contrary, the lower the oxygen concentration of the metal oxide layer **130**, the lower the resistivity of the metal oxide layer **130**. In other words, the thickness of the oxygen-containing protrusive structure **122** affects the oxygen concentration of the metal oxide layer **130** and further poses an impact on its resistivity. For instance, the first portion **132** of the metal oxide layer **130** covers a central portion of the oxygen-containing protrusive structure **122** of a relatively large thickness, and most of the second portion **134** of the metal oxide layer **130** covers an edge portion of the oxygen-containing protrusive structure **122** of a gradually reduced thickness. Therefore, the oxygen concentration of the first portion **132** is higher than the oxygen concentration of the second portion **134**, and the resistivity of the first portion **132** is higher than the resistivity of the second portion **134**. Since the thickness of the edge portion of the oxygen-containing protrusive structure **122** covered by the second portion **134** gradually decreases in a direction away from the first portion **132**, the oxygen concentration of the second portion **134** also gradually decreases in the direction away from the first portion **132**, so that the resistivity of the second portion **134** gradually decreases away from the first portion **132**. As such, the issue of a lateral electric field generated due to a sudden change to the resistivity of the metal oxide layer **130** may be solved to a great extent by arranging the second portion **134**, thereby improving the reliability of the semiconductor device **1**.

(16) In this embodiment, since a width L1 of the first gate **150** is slightly smaller than a width of the first surface **122***a* of the oxygen-containing protrusive structure **122**, the second portion **134** in the doped region dp partially covers a part of the first surface **122***a* of the oxygen-containing protrusive structure **122**, which should however not be construed as a limitation in the disclosure. In other embodiments, since the width L1 of the first gate **150** is slightly larger than or equal to the width of the first surface **122***a* of the oxygen-containing protrusive structure **122**, the second portion **134** in the doped region dp does not cover the first surface **122***a* of the oxygen-containing protrusive structure **122**.

(17) The third portion **136** of the metal oxide layer **130** is not in contact with the oxygen-containing protrusive structure **122**, and therefore the third portion **136** has a lower oxygen concentration than that of the first portion **132** and that of the second portion **134**. That is, the resistivity of the first portion **132** and the resistivity of the second portion **134** are both higher than the resistivity of the third portion **136**.

(18) The interlayer dielectric layer **160** is disposed on the gate dielectric layer **140** and covers the first gate **150**. A material of the interlayer dielectric layer **160** and the gate dielectric layer **140** includes, for instance, silicon oxide, silicon nitride, SiNO, or other appropriate materials. The source **172** and the drain **174** are located on the interlayer dielectric layer **160** and penetrate the interlayer dielectric layer **160** and the gate dielectric layer **140**, so as to be electrically connected to the third portion **136** of the metal oxide layer **130**. Since the resistivity of the third portion **136** is smaller than the resistivity of the second portion **134**, an interface resistance between the source **172** and the third portion **136** and an interface resistance between the drain **174** and the third portion **136** may be reduced, thereby increasing an operating current of the semiconductor device **1**.

(19) For the purpose of clarity, in FIG. **1**, a clear boundary line is illustrated between the first portion **132** and the second portion **134** and between the second portion **134** and the third portion **136** of the metal oxide layer **130**, but it should be understood that these boundary lines are only provided for explanation. In the actual structure, there may be no such clear boundary lines because of the gradual changes between the first portion **132** and the second portion **134** and between the second portion **134** and the third portion **136**.

(20) FIG. **2**A to FIG. **2**E are schematic cross-sectional flowcharts of a manufacturing process of the semiconductor device depicted in FIG. **1**.

(21) With reference to FIG. **2**A, a substrate **100** is provided, a buffer layer **110** is formed on the substrate **100**, and an oxygen-containing material layer **120** is deposited on the substrate **100** and the buffer layer **110**.

(22) With reference to FIG. 2B, the oxygen-containing material layer **120** is patterned by performing a dry etching process or a wet etching process to expose a partial surface of the buffer layer **110**, whereby an oxygen-containing protrusive structure **122** is formed. The oxygen-containing protrusive structure **122** is, for instance, a trapezoidal structure and has a first surface **122***a*, a second surface **122***b* opposite to the first surface **122***a*, and a plurality of sidewalls **122***c* connected to the first surface **122***a* and the second surface **122***b*.
(23) With reference to FIG. 2C, a metal oxide layer **130′** is conformally formed on the oxygen-containing protrusive structure **122** and the buffer layer **110**. The metal oxide layer **130′** may include a first portion **132**, a second portion **134**, and a third portion **136**. The first portion **132** covers the first surface **122***a* of the oxygen-containing protrusive structure **122**. The second portion **134** is connected to the first portion **132** and covers the sidewalls **122***c* of the oxygen-containing protrusive structure **122**. The third portion **136** is connected to the second portion **134** and extends from the sidewalls **122***c* of the oxygen-containing protrusive structure **122** in a direction away from the oxygen-containing protrusive structure **122**.
(24) With reference to FIG. 2D, a gate dielectric layer **140** is formed on the metal oxide layer **130′**, and a first gate **150** is formed on the gate dielectric layer **140**, where the first gate **150** is overlapped with the metal oxide layer **130′** in a normal direction ND of a top surface of the substrate **100**. For instance, the first gate **150** may be overlapped with the first portion **132** in the normal direction ND of the top surface of the substrate **100**.
(25) In some embodiments, before the gate dielectric layer **140** and the first gate **150** are formed, an annealing process may be performed on the metal oxide layer **130′** or the gate dielectric layer **140** to adjust an oxygen distribution in the metal oxide layer **130′** by the oxygen-containing protrusive structure **122**. For instance, the first portion **132** and the second portion **134** of the metal oxide layer **130′** are located on the oxygen-containing protrusive structure **122**, and thus the oxygen-containing protrusive structure **122** may provide oxygen to the first portion **132** and the second portion **134** in the annealing process; the third portion **136** is not formed on the oxygen-containing protrusive structure **122**, and thus the oxygen in the third portion **136** may be easily dissipated, so that the oxygen concentration of the third portion **136** is lower than the oxygen concentration of the first portion **132** and the oxygen concentration of the second portion **134**. In other embodiments, the annealing process is performed on the metal oxide layer **130′** or the gate dielectric layer **140** after the gate dielectric layer **140** is formed and before the first gate **150** is formed to adjust the oxygen distribution in the metal oxide layer **130′** by the oxygen-containing protrusive structure **122**. For instance, the oxygen-containing protrusive structure **122** may provide oxygen to the first portion **132** and the second portion **134** in the annealing process, so as to increase the oxygen concentration of the first portion **132** and the oxygen concentration of the second portion **134**, which allows the oxygen concentration of the first portion **132** and the oxygen concentration of the second portion **134** to be higher than the oxygen concentration of the third portion **136**.
(26) With reference to FIG. 2D, a doping process P is performed on the metal oxide layer **130** to reduce the resistivity of the third portion **136** of the metal oxide layer **130** and gradually decrease the resistivity of the second portion **134** of the metal oxide layer **130** away from the first portion **132**. For instance, a hydrogen plasma process is performed on the metal oxide layer **130** by applying the first gate **150** as a mask, so that the second portion **134** and the third portion **136** that are not overlapped with the first gate **150** in the normal direction ND of the top surface of the substrate **100** constitute a doped region dp, and the first portion **132** overlapped with the first gate **150** in the normal direction ND of the top surface of the substrate **100** constitutes a channel region ch. In some embodiments, the second portion **134** may constitute a lightly doped region ldp, and the third portion **136** may constitute a heavily doped region hdp. In some embodiments, a width L2 of the oxygen-containing protrusive structure **122** may be larger than a width L1 of the first gate **150**, which is conducive to an increase in the distance between the first portion **132** and the third portion **136**.

(27) During the hydrogen plasma process, the oxygen in the metal oxide layer **130** may react with hydrogen to create oxygen vacancies in the metal oxide layer **130**, thereby reducing the resistivity of the metal oxide layer **130**. Besides, the thickness of the oxygen-containing protrusive structure **122** covered by the second portion **134** gradually decreases, and the third portion **136** is not in contact with the oxygen-containing protrusive structure **122**; therefore, the oxygen concentration of the second portion **134** gradually decreases in the direction away from the first portion **132**, and the oxygen concentration of the second portion **134** is higher than the oxygen concentration of the third portion **136**. As a result, after the hydrogen plasma process, the resistivity of the second portion **134** may gradually decrease from the first portion **132**, and the resistivity of the second portion **134** is higher than the resistivity of the third portion **136**.
(28) With reference to FIG. **2**E, an interlayer dielectric layer **160** is formed on the gate dielectric layer **140** and covers the first gate **150**. After that, openings O1 and O2 penetrating the interlayer dielectric layer **160** and the gate dielectric layer **140** are formed, and the openings O1 and O2 are respectively overlapped with the third portion **136** in the normal direction ND of the top surface of the substrate **100**.
(29) After that, with reference to FIG. **1**, a source **172** and a drain **174** are formed on the interlayer dielectric layer **150** and fill the openings O1 and O2, so as to be electrically connected to the metal oxide layer **130**.
(30) After the above process, the fabrication of the semiconductor device **1** is substantially completed.
(31) FIG. **3** is a schematic cross-sectional view of a semiconductor device according to another embodiment of the disclosure. Note that the reference numbers and some contents provided in the embodiment depicted in FIG. **1** are used in the embodiment depicted in FIG. **3**, where the same or similar numbers are applied to denote the same or similar elements, and the description of the same technical content is omitted. The description of the omitted content may be found in the previous embodiment and will not be provided hereinafter.
(32) With reference to FIG. **3**, the main difference between a semiconductor device **2** shown in FIG. **3** and the semiconductor device **1** shown in FIG. **1** lies in that the semiconductor device **2** further includes an extension structure **124**. The extension structure **124** is located between the third portion **136** of the metal oxide layer **130** and the substrate **100** and is connected to the oxygen-containing protrusive structure **122** integrally, and a thickness T1 of the oxygen-containing protrusive structure **122** between the first surface **122***a* and the second surface **122***b* is larger than a thickness T2 of the extension structure **124**. A material of the extension structure **124** is the same as the material of the oxygen-containing protrusive structure **122**. The thickness of the oxygen-containing protrusive structure **122** gradually decreases with proximity to the edge, and the thickness T1 of the oxygen-containing protrusive structure **122** between the first surface **122***a* and the second surface **122***b* is larger than the thickness T2 of the extension structure **124**; hence, the oxygen concentration of the first portion **132** is higher than the oxygen concentration of the second portion **134**, and the oxygen concentration of the second portion **134** is higher than the oxygen concentration of the third portion **136**. Thereby, the resistivity of the first portion **132** is higher than the resistivity of the second portion **134**, and the resistivity of the second portion **134** is higher than the resistivity of the third portion **136**. In addition, since the thickness of a part of the oxygen-containing protrusive structure **122** covered by the second portion **134** gradually decreases with proximity to the edge, the oxygen concentration of the second portion **134** gradually decreases in the direction away from the first portion **132**, so that the resistivity of the second portion **134** gradually decreases in the direction away from the first portion **134**. As such, the issue of a lateral electric field generated due to a sudden change to the resistivity of the metal oxide layer **130** may be solved to a great extent by arranging the second portion **134**, thereby improving the reliability of the semiconductor device **2**. Besides, since the resistivity of the third portion **136** is smaller than the resistivity of the second portion **134**, the interface resistance between the source **172** and the

third portion **136** and the interface resistance between the drain **174** and the third portion **136** may be reduced, thereby increasing an operating current of the semiconductor device **2**.
(33) FIG. **4**A to FIG. **4**D are schematic cross-sectional flowcharts of a manufacturing process of the semiconductor device depicted in FIG. **3**. Here, FIG. **4**A to FIG. **4**D may be considered as schematic cross-sectional views of the manufacturing method of the semiconductor device following the steps depicted in FIG. **2**A. The description of the steps in FIG. **2**A may be found in the previous embodiment and will not be provided hereinafter.
(34) With reference to FIG. **4**A, an oxygen-containing material layer **120** is patterned by performing a dry etching process or a wet etching process, so as to remove a part of the oxygen-containing material layer **120** without exposing the surface of a buffer layer **110** covered by the oxygen-containing material layer **120**, so as to form an oxygen-containing structure **122** and an extension structure **124**. The oxygen-containing protrusive structure **122** is, for instance, a trapezoidal structure and has a first surface **122***a*, a second surface **122***b* opposite to the first surface **122***a*, and a plurality of sidewalls **122***c* connected to the first surface **122***a* and the second surface **122***b*. The extension structure **124** is integrally connected to the oxygen-containing protrusive structure **122**, and a thickness T1 of the oxygen-containing protrusive structure **122** between the first surface **122***a* and the second surface **122***b* is larger than a thickness T2 of the extension structure **124**.
(35) With reference to FIG. **4**B, a metal oxide layer **130′** is conformally formed on the oxygen-containing protrusive structure **122** and an extension structure **124**. The metal oxide layer **130′** may include a first portion **132**, a second portion **134**, and a third portion **136**. The first portion **132** covers the first surface **122***a* of the oxygen-containing protrusive structure **122**. The second portion **134** is connected to the first portion **132** and covers the sidewalls **122***c* of the oxygen-containing protrusive structure **122**. The third portion **136** is connected to the second portion **134** and extends from the sidewalls **122***c* of the oxygen-containing protrusive structure **122** in a direction away from the oxygen-containing protrusive structure **122**. The extension structure **124** is located between the third portion **136** and the buffer layer **110**.
(36) With reference to FIG. **4**C, a gate dielectric layer **140** is formed on the metal oxide layer **130′**, and a first gate **150** is formed on the gate dielectric layer **140**, where the first gate **150** is overlapped with the metal oxide layer **130′** in a normal direction ND of a top surface of the substrate **100**. For instance, the first gate **150** may be overlapped with the first portion **132** in the normal direction ND of the top surface of the substrate **100**.
(37) In some embodiments, before the gate dielectric layer **140** and the first gate **150** are formed, an annealing process may be performed on the metal oxide layer **130′** or the gate dielectric layer **140** to adjust an oxygen distribution in the metal oxide layer **130′** by the oxygen-containing protrusive structure **122**. In other embodiments, the annealing process is performed on the metal oxide layer **130′** or the gate dielectric layer **140** after the gate dielectric layer **140** is formed and before the first gate **150** is formed to adjust the oxygen distribution in the metal oxide layer **130′** by the oxygen-containing protrusive structure **122**.
(38) With reference to FIG. **4**C, a doping process P is performed on the metal oxide layer **130** to reduce the resistivity of the third portion **136** of the metal oxide layer **130** and gradually decrease the resistivity of the second portion **134** of the metal oxide layer **130** away from the first portion **132**. For instance, a hydrogen plasma process is performed on the metal oxide layer **130** by applying the first gate **150** as a mask, so that the second portion **134** and the third portion **136** that are not overlapped with the first gate **150** in the normal direction ND of the top surface of the substrate **100** constitute a doped region dp, and the first portion **132** overlapped with the first gate **150** in the normal direction ND of the top surface of the substrate **100** constitutes a channel region ch. In some embodiments, the second portion **134** may constitute a lightly doped region ldp, and the third portion **136** may constitute a heavily doped region hdp. In some embodiments, a width L2 of the oxygen-containing protrusive structure **122** may be larger than a width L1 of the first gate

150, which is conducive to an increase in the distance between the first portion **132** and the third portion **136**.
(39) After the hydrogen plasma process, the resistivity of the second portion **134** may gradually decrease away from the first portion **132**, and the resistivity of the second portion **134** is higher than the resistivity of the third portion **136**.
(40) With reference to FIG. **4**D, an interlayer dielectric layer **160** is formed on the gate dielectric layer **140** and covers the first gate **150**. After that, openings O1 and O2 penetrating the interlayer dielectric layer **160** and the gate dielectric layer **140** are formed, and the openings O1 and O2 are respectively overlapped with the third portion **136** in the normal direction ND of the top surface of the substrate **100**.
(41) After that, with reference to FIG. **3**, a source **172** and a drain **174** are formed on the interlayer dielectric layer **150** and fill the openings O1 and O2, so as to be electrically connected to the metal oxide layer **130**.
(42) After the above process, the fabrication of the semiconductor device **2** is substantially completed.
(43) FIG. **5** is a schematic cross-sectional view of a semiconductor device according to still another embodiment of the disclosure. Note that the reference numbers and some contents provided in the embodiment depicted in FIG. **1** are used in the embodiment depicted in FIG. **5**, where the same or similar numbers are applied to denote the same or similar elements, and the description of the same technical content is omitted. The description of the omitted content may be found in the previous embodiment and will not be provided hereinafter.
(44) With reference to FIG. **5**, the main difference between a semiconductor device **3** shown in FIG. **5** and the semiconductor device **1** shown in FIG. **1** lies in that the semiconductor device **3** further includes a second gate **180** which may be disposed between the substrate **100** and the oxygen-containing protrusive structure **122**. For instance, the second gate **180** is disposed between the substrate **100** and the buffer layer **110**. That is, the semiconductor device **3** is a dual-gate transistor, and thus the performance and the stability of the semiconductor device **3** may be improved. A width L3 of the second gate **180** may be larger than the width L2 of the oxygen-containing protrusive structure **122**, so as to separate the electric field between the gate and the drain in the semiconductor device **3**, thereby improving the reliability of the semiconductor device **3**.
(45) FIG. **6**A to FIG. **6**D are schematic cross-sectional flowcharts of a manufacturing process of the semiconductor device depicted in FIG. **5**.
(46) With reference to FIG. **6**A, a substrate **100** is provided, a second gate **180** is formed on the substrate **100**, and a buffer layer **110** is formed on the second gate **180** and the substrate **100** and covers the second gate **180**.
(47) With reference to FIG. **6**B, an oxygen-containing protrusive structure **122** is formed on the buffer layer **110**. The oxygen-containing protrusive structure **122** is, for instance, a trapezoidal structure and has a first surface **122***a*, a second surface **122***b* opposite to the first surface **122***a*, and a plurality of sidewalls **122***c* connected to the first surface **122***a* and the second surface **122***b*. After that, a metal oxide layer **130**′ is conformally formed on the oxygen-containing protrusive structure **122** and the buffer layer **110**. The metal oxide layer **130**′ may include a first portion **132**, a second portion **134**, and a third portion **136**. The first portion **132** covers the first surface **122***a* of the oxygen-containing protrusive structure **122**. The second portion **134** is connected to the first portion **132** and covers the sidewalls **122***c* of the oxygen-containing protrusive structure **122**. The third portion **136** is connected to the second portion **134** and extends from the sidewalls **122***c* of the oxygen-containing protrusive structure **122** in a direction away from the oxygen-containing protrusive structure **122**. In an embodiment, the width L3 of the second gate **180** may be larger than the width L2 of the oxygen-containing protrusive structure **122**.
(48) With reference to FIG. **6**C, a gate dielectric layer **140** is formed on the metal oxide layer **130**′,

and a first gate **150** is formed on the gate dielectric layer **140**, where the first gate **150** is overlapped with the metal oxide layer **130**′ in a normal direction ND of a top surface of the substrate **100**. For instance, the first gate **150** may be overlapped with the first portion **132** in the normal direction ND of the top surface of the substrate **100**.

(49) In some embodiments, before the gate dielectric layer **140** and the first gate **150** are formed, an annealing process may be performed on the metal oxide layer **130**′ or the gate dielectric layer **140** to adjust an oxygen distribution in the metal oxide layer **130**′ by the oxygen-containing protrusive structure **122**. In other embodiments, the annealing process is performed on the metal oxide layer **130**′ or the gate dielectric layer **140** after the gate dielectric layer **140** is formed and before the first gate **150** is formed to adjust the oxygen distribution in the metal oxide layer **130**′ by the oxygen-containing protrusive structure **122**.

(50) With reference to FIG. **6**C, a doping process P is performed on the metal oxide layer **130** to reduce the resistivity of the third portion **136** of the metal oxide layer **130** and gradually decrease the resistivity of the second portion **134** of the metal oxide layer **130** away from the first portion **132**. For instance, a hydrogen plasma process is performed on the metal oxide layer **130** by applying the first gate **150** as a mask, so that the second portion **134** and the third portion **136** that are not overlapped with the first gate **150** in the normal direction ND of the top surface of the substrate **100** constitute a doped region dp, and the first portion **132** overlapped with the first gate **150** in the normal direction ND of the top surface of the substrate **100** constitutes a channel region ch. In some embodiments, the second portion **134** may constitute a lightly doped region ldp, and the third portion **136** may constitute a heavily doped region hdp. In some embodiments, the width L2 of the oxygen-containing protrusive structure **122** may be larger than the width L1 of the first gate **150**, which is conducive to an increase in the distance between the first portion **132** and the third portion **136**.

(51) After the hydrogen plasma process, the resistivity of the second portion **134** may gradually decrease away from the first portion **132**, and the resistivity of the second portion **134** is higher than the resistivity of the third portion **136**.

(52) With reference to FIG. **6**D, an interlayer dielectric layer **160** is formed on the gate dielectric layer **140** and covers the first gate **150**. After that, openings O1 and O2 penetrating the interlayer dielectric layer **160** and the gate dielectric layer **140** are formed, and the openings O1 and O2 are respectively overlapped with the third portion **136** in the normal direction ND of the top surface of the substrate **100**.

(53) After that, with reference to FIG. **5**, a source **172** and a drain **174** are formed on the interlayer dielectric layer **150** and fill the openings O1 and O2, so as to be electrically connected to the metal oxide layer **130**.

(54) After the above process, the fabrication of the semiconductor device **3** is substantially completed.

(55) It will be apparent to those skilled in the art that various modifications and variations can be made to the disclosed embodiments without departing from the scope or spirit of the disclosure. In view of the foregoing, it is intended that the disclosure covers modifications and variations provided that they fall within the scope of the following claims and their equivalents.

Claims

1. A semiconductor device, comprising: a substrate; an oxygen-containing protrusive structure, disposed above the substrate and having a first surface, a second surface opposite to the first surface, and a sidewall connected to the first surface and the second surface; a buffer layer, disposed between the substrate and the oxygen-containing protrusive structure, wherein a material of the buffer layer is silicon nitride or silicon nitride oxide, and an oxygen concentration of the buffer layer is lower than an oxygen concentration of the oxygen-containing protrusive structure; a

metal oxide layer, comprising: a first portion, covering the first surface of the oxygen-containing protrusive structure; a second portion, connected to the first portion and covering at least a portion of the sidewall of the oxygen-containing protrusive structure, wherein a resistivity of the second portion gradually decreases away from the first portion; and a third portion, connected to the second portion and extending from the second portion in a direction away from the oxygen-containing protrusive structure; a gate dielectric layer, disposed on the metal oxide layer; a first gate, disposed on the gate dielectric layer; a source and a drain, disposed on the metal oxide layer, wherein the first gate is located between the source and the drain; and a second gate, disposed between the substrate and the buffer layer, wherein a width of the second gate is larger than a width of the oxygen-containing protrusive structure, wherein an orthographic projection of the second gate on the substrate has a first point adjacent to the source and a second point adjacent to the drain, and the width of the second gate is a distance between the first point and the second point, and wherein an orthographic projection of the oxygen-containing protrusive structure on the substrate has a third point adjacent to the source and a fourth point adjacent to the drain, and the width of the oxygen-containing protrusive structure is a distance between the third point and the fourth point, wherein the gate dielectric layer contacts the buffer layer, the source, the drain, the first portion of the metal oxide layer, the second portion of the metal oxide layer, and the third portion of the metal oxide layer, wherein the third portion of the metal oxide layer is in direct contact with the buffer layer but not in direct contact with the oxygen-containing protrusive structure, wherein the oxygen-containing protrusive structure has a gradually reduced thickness at an edge portion of the oxygen-containing protrusive structure, and an oxygen concentration of the second portion is gradually decreased along the gradually reduced thickness of the oxygen-containing protrusive structure.

2. The semiconductor device according to claim 1, further comprising an extension structure, wherein the extension structure is located between the third portion and the substrate and is connected to the oxygen-containing protrusive structure integrally, and a thickness of the oxygen-containing protrusive structure is larger than a thickness of the extension structure.

3. The semiconductor device according to claim 1, wherein a resistivity of the first portion is higher than the resistivity of the second portion, and the resistivity of the second portion is higher than a resistivity of the third portion.

4. The semiconductor device according to claim 1, wherein an oxygen concentration of the first portion is higher than the oxygen concentration of the second portion, and the oxygen concentration of the second portion is higher than an oxygen concentration of the third portion.

5. The semiconductor device according to claim 1, wherein a material of the oxygen-containing protrusive structure comprises silicon oxide or silicon nitride oxide.

6. The semiconductor device according to claim 1, wherein the first gate is overlapped with the first portion in a normal direction of a top surface of the substrate, and the first gate is not overlapped with the second portion in the normal direction of the top surface of the substrate.

7. The semiconductor device according to claim 1, wherein the width of the oxygen-containing protrusive structure is larger than a width of the first gate.

8. A manufacturing method of a semiconductor device, comprising: providing a substrate; forming a second gate on the substrate; forming a buffer layer on the substrate and the second gate, wherein a material of the buffer layer is silicon nitride or silicon nitride oxide; forming an oxygen-containing protrusive structure above the buffer layer, wherein the oxygen-containing protrusive structure has a first surface, a second surface opposite to the first surface, and a sidewall connected to the first surface and the second surface, wherein an oxygen concentration of the buffer layer is lower than an oxygen concentration of the oxygen-containing protrusive structure; forming a metal oxide layer on the oxygen-containing protrusive structure, wherein the metal oxide layer comprises: a first portion, covering the first surface of the oxygen-containing protrusive structure; a second portion, connected to the first portion and covering at least a portion of the sidewall of the oxygen-containing protrusive structure; and a third portion, connected to the second portion and

extending from the second portion in a direction away from the oxygen-containing protrusive structure; forming a gate dielectric layer on the metal oxide layer; forming a first gate on the gate dielectric layer, wherein the first gate is overlapped with the metal oxide layer in a normal direction of a top surface of the substrate; and performing a doping process on the metal oxide layer to reduce a resistivity of the third portion of the metal oxide layer and gradually decreasing a resistivity of the second portion of the metal oxide layer away from the first portion; and forming a source and a drain on the metal oxide layer, wherein the first gate is located between the source and the drain, wherein a width of the second gate is larger than a width of the oxygen-containing protrusive structure, wherein an orthographic projection of the second gate on the substrate has a first point adjacent to the source and a second point adjacent to the drain, and the width of the second gate is a distance between the first point and the second point, and wherein an orthographic projection of the oxygen-containing protrusive structure on the substrate has a third point adjacent to the source and a fourth point adjacent to the drain, and the width of the oxygen-containing protrusive structure is a distance between the third point and the fourth point, wherein the gate dielectric layer contacts the buffer layer, the source, the drain, the first portion of the metal oxide layer, the second portion of the metal oxide layer, and the third portion of the metal oxide layer, wherein the third portion of the metal oxide layer is in direct contact with the buffer layer but not in direct contact with the oxygen-containing protrusive structure, wherein the oxygen-containing protrusive structure has a gradually reduced thickness at an edge portion of the oxygen-containing protrusive structure, and an oxygen concentration of the second portion is gradually decreased along the gradually reduced thickness of the oxygen-containing protrusive structure.

9. The manufacturing method according to claim 8, wherein the step of forming the oxygen-containing protrusive structure above the substrate comprises: depositing an oxygen-containing material layer on the substrate; and patterning the oxygen-containing material layer to form the oxygen-containing protrusive structure.

10. The manufacturing method according to claim 9, wherein the oxygen-containing material layer is patterned to form the oxygen-containing protrusive structure and an extension structure, the extension structure is connected to the oxygen-containing protrusive structure integrally, and a thickness of the oxygen-containing protrusive structure is larger than a thickness of the extension structure.

11. The semiconductor device according to claim 1, wherein the third portion of the metal oxide layer contacts and covers a portion of the buffer layer, and a thickness of the portion of the buffer layer is smaller than a thickness of the second gate.

12. The semiconductor device according to claim 1, wherein the third portion of the metal oxide layer contacts and covers a portion of the buffer layer, and wherein a distance between a top surface of the portion of the buffer layer and a top surface of the substrate is smaller than a distance between a top surface of the second gate and the top surface of the substrate.